Year 11 Set 4 · Foundation · Lesson 5

Pythagoras' theorem

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Today I will learn to...

Goals

1. Find the hypotenuse using Pythagoras' theorem

2. Find a shorter side using Pythagoras' theorem

Do now: retrieval practice

4 mins

1 Area of a circle with radius 2, in terms of π

4π

2 Work out √81

9

3 Work out 7^2

49

4 Solve x + 12 = 20

x = 8

Objective 1

Find the hypotenuse using Pythagoras' theorem

Key idea

Learn

!

$a^2 + b^2 = c^2$, where c is the hypotenuse (the longest side, opposite the right angle). Square, ADD, square root.

Objective 1: Find the hypotenuse using Pythagoras' theorem

I do: example 1

I do

Shorter sides 3 cm and 4 cm

1. Finding the hypotenuse: $c^2 = a^2 + b^2$
2. $c^2 = 3^2 + 4^2 = 9 + 16$
3. $c^2 = 25$
4. $c = \sqrt{25} = 5$ cm

I do: example 2

I do

Shorter sides 6 cm and 8 cm

1. Finding the hypotenuse: $c^2 = a^2 + b^2$
2. $c^2 = 6^2 + 8^2 = 36 + 64$
3. $c^2 = 100$
4. $c = \sqrt{100} = 10$ cm

I do: example 3

I do

Shorter sides 5 cm and 7 cm

1. Finding the hypotenuse: $c^2 = a^2 + b^2$
2. $c^2 = 5^2 + 7^2 = 25 + 49$
3. $c^2 = 74$
4. $c = \sqrt{74} = 8.60$ cm (2 dp)

Mini whiteboards

We do

1. Sides 5 and 12. Hypotenuse?

Answer: 13

2. Sides 2 and 3. Hypotenuse (2 dp)?

Answer: 3.61

3. Sides 6 and 8. Hypotenuse?

Answer: 10

4. Sides 4 and 7. Hypotenuse (1 dp)?

Answer: 8.1

You do: choose your challenge

You do

Bronze

Shorter sides 8 cm and 15 cm. Find the hypotenuse.

Silver

Shorter sides 7 cm and 9 cm. Find the hypotenuse (2 dp).

Gold

A rectangle is 12 cm by 5 cm. Find the length of its diagonal.

Platinum

Tia walks 6 km north then 9 km east. How far is she from her start (1 dp)?

You do: answers

Check

Bronze	17 cm
Silver	√130 = 11.40 cm
Gold	13 cm
Platinum	√117 = 10.8 km

If you finish early

Extra

Bronze

Shorter sides 9 cm and 12 cm. Find the hypotenuse.

15 cm

Silver

Shorter sides 5 cm and 9 cm. Find the hypotenuse (1 dp).

√106 = 10.3 cm

Gold

A square has side 7 cm. Find the length of its diagonal (1 dp).

√98 = 9.9 cm

Platinum

Find the distance between the points (1, 2) and (7, 10).

√(36 + 64) = 10

Exam question 1

2 marks

A right-angled triangle has shorter sides 9 cm and 12 cm. Work out the hypotenuse.

Space for working

Mark scheme: 81 + 144 = 225, √225 = 15 cm

Exam question 2

3 marks

A ladder's foot is 1.5 m from a wall and its top is 4 m up the wall. How long is the ladder? (1 dp)

Space for working

Mark scheme: 2.25 + 16 = 18.25, √18.25 = 4.3 m

Objective 1 achieved

Find the hypotenuse using Pythagoras' theorem



Objective 2

Find a shorter side using Pythagoras' theorem

✓ Achieved: Find the hypotenuse using Pythagoras' theorem

Key idea

Learn

!

For a shorter side: square, SUBTRACT, square root.
Hypotenuse² − other side².

Objective 2: Find a shorter side using Pythagoras' theorem

I do: example 1

I do

Hypotenuse 13 cm, one side 5 cm

1. Finding a shorter side: $a^2 = c^2 - b^2$
2. $a^2 = 13^2 - 5^2 = 169 - 25$
3. $a^2 = 144$
4. $a = \sqrt{144} = 12$ cm

I do: example 2

I do

Hypotenuse 10 cm, one side 6 cm

1. Finding a shorter side: $a^2 = c^2 - b^2$
2. $a^2 = 10^2 - 6^2 = 100 - 36$
3. $a^2 = 64$
4. $a = \sqrt{64} = 8$ cm

I do: example 3

I do

Hypotenuse 9 cm, one side 4 cm

1. Finding a shorter side: $a^2 = c^2 - b^2$
2. $a^2 = 9^2 - 4^2 = 81 - 16$
3. $a^2 = 65$
4. $a = \sqrt{65} = 8.06$ cm (2 dp)

Mini whiteboards

We do

1. Hypotenuse 17, side 8. Other side?

Answer: 15

2. Hypotenuse 7, side 3 (2 dp)?

Answer: 6.32

3. Hypotenuse 10, side 6. Other side?

Answer: 8

4. Hypotenuse 9, side 4 (1 dp)?

Answer: 8.1

You do: choose your challenge

You do

Bronze

Hypotenuse 25 cm, one side 7 cm. Find the other side.

Silver

Hypotenuse 12 cm, one side 5 cm. Find the other side (2 dp).

Gold

A TV is 50 inches on the diagonal and 40 inches wide. How tall is it?

Platinum

An isosceles triangle has sides 13 cm, 13 cm and 10 cm. Find its area.

You do: answers

Check

Bronze

24 cm

Silver

$\sqrt{119} = 10.91$ cm

Gold

30 inches

Platinum

Height 12 cm, area = 60 cm^2

If you finish early

Extra

Bronze

Hypotenuse 13 cm, one side 5 cm. Find the other side.

12 cm

Silver

Hypotenuse 11 cm, one side 6 cm. Find the other side (1 dp).

√85 = 9.2 cm

Gold

A 5 m ladder leans on a wall with its foot 1.4 m from the wall. How high up the wall does it reach?

√(25 − 1.96) = 4.8 m

Platinum

An isosceles triangle has sides 10 cm, 10 cm and 12 cm. Find its height and area.

Height 8 cm, area 48 cm²

Exam question 1

3 marks

A 5 m ladder leans against a wall with its foot 1.4 m from the wall. How high up the wall does it reach? (1 dp)

Space for working

Mark scheme: 25 − 1.96 = 23.04, √23.04 = 4.8 m

Exam question 2

2 marks

Show that a triangle with sides 7 cm, 24 cm and 25 cm is right-angled.

Space for working

Mark scheme: $7^2 + 24^2 = 49 + 576 = 625 = 25^2$ ✓

Objective 2 achieved

Find a shorter side using Pythagoras' theorem



Real life

Apply

A TV screen is 32 inches wide and 18 inches high. TVs are sold by diagonal length. What is its size? (1 dp)

$\sqrt{(32^2 + 18^2)} = \sqrt{1348} = 36.7$ inches

Extension: stretch yourself

Stretch

1. **Find the distance between (1, 2) and (7, 10).**

$\sqrt{(6^2 + 8^2)} = 10$

2. **An isosceles triangle has sides 10, 10 and 12 cm. Find its area.**

Height = $\sqrt{(100 - 36)} = 8$, area = $\frac{1}{2} \times 12 \times 8 = 48\ \text{cm}^2$

Exit ticket: exam questions

Exit

1. The shorter sides of a right-angled triangle are 6 cm and 8 cm. Find the hypotenuse. **(2 marks)**

2. The hypotenuse is 15 cm and one side is 9 cm. Find the other side. **(2 marks)**

3. A 6 m ladder has its foot 2 m from a wall. How high up the wall does it reach? (1 dp) **(3 marks)**

Exit ticket: mark scheme

Mark it

Q1 10 cm [2]

Q2 12 cm [2]

Q3 $\sqrt{(36 - 4)} = \sqrt{32} = 5.7$ m [3]

Total: ____ / 7

Today I achieved

- ✔ **Find the hypotenuse using Pythagoras' theorem**
- ✔ **Find a shorter side using Pythagoras' theorem**

Rate yourself: 1 = still unsure 2 = getting there 3 = confident

Homework: exam questions

Due next lesson

1. Shorter sides 9 cm and 40 cm. Find the hypotenuse. [2]
2. Shorter sides 4 cm and 6 cm. Find the hypotenuse to 1 dp. [2]
3. Hypotenuse 20 cm, one side 16 cm. Find the other side. [2]
4. A rectangle has a diagonal of 10 cm and a width of 6 cm. Find its length and its perimeter. [3]
5. Find the distance between the points (2, 1) and (8, 9). [3]

Extra practice: Maths Genie → Edexcel question bank → Pythagoras